Graph Sketching: Quadratic Rational Functions

Questions 1–25

Question 1. Find any asymptotes for each of the graphs of the curves below:

a) $y = \dfrac{x^2 - 1}{x^2 + 3}$

b) $y = \dfrac{x^2 - 16}{x^2 + 7}$

c) $y = \dfrac{x^2 + 3}{x^2 - 2}$

d) $y = \dfrac{3x^2 - 5x + 2}{x^2 + 3x}$

Question 2. Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly. It is given that each curve has a stationary point when $x = 0$

a) $y = \dfrac{x^2 - 4}{x^2 + 9}$

b) $y = \dfrac{x^2 - 9}{x^2 + 4}$

c) $y = \dfrac{x^2}{x^2 + 1}$

Question 3. Solve the following equations:

a) $\dfrac{2x^2-3x+23}{2x^2+3x+9}=2$

b) $\dfrac{x^2-4x+2}{x^2-2x-5}=\dfrac{1}{2}$

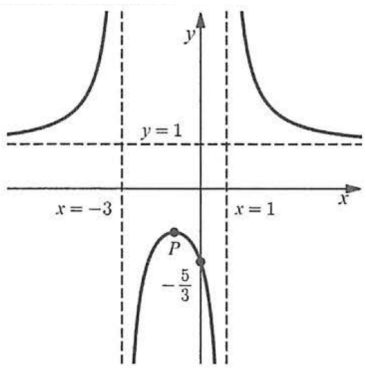
c) $\dfrac{x^2-4x+3}{x^2-3x+2}=\dfrac{4}{3}$

d) $\dfrac{x^2+4x-5}{x^2+4}=-\dfrac{1}{3}$

Question 4. The curve

$$y=\frac{x^2+2x+5}{x^2+2x-3}$$

is shown in the diagram below. The local maximum is labelled P



a) By considering where the line $y=k$ intersects the curve, show that

$$(k-1)(k+1)\geq 0$$

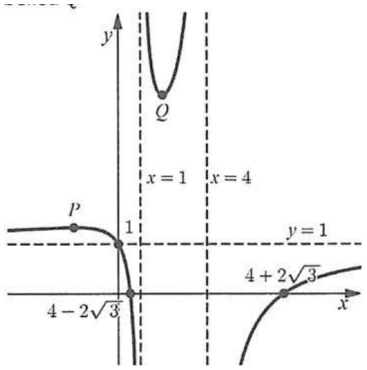
b) State the values that y can be.

c) Hence, find the y-coordinate of the point P

Question 5. The curve

$$y = \frac{x^2 - 8x + 4}{x^2 - 5x + 4}$$

is shown in the diagram below. The local maximum is labelled P and the local minimum is labelled Q



a) By considering where the line $y = k$ intersects the curve, show that

$$9k^2 - 48k + 48 \geq 0$$

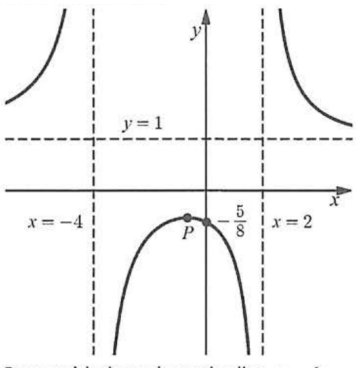
b) State the values that y can be.

c) Hence, find the y-coordinate of the point P and the y-coordinate of the point Q

Question 6. The curve

$$y = \frac{x^2 + x + 5}{x^2 + 2x - 8}$$

is shown in the diagram below. The local maximum is labelled P



a) By considering where the line $y = k$ intersects the curve, show that

$$36k^2 - 16k - 19 \geq 0$$

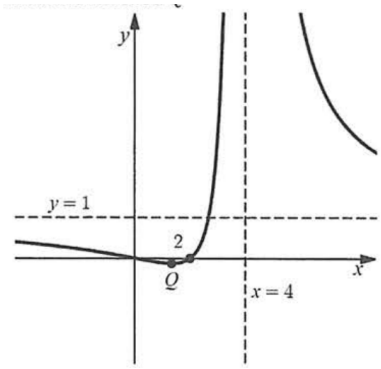
b) State the values that y can be.

c) Hence, find the y-coordinate of the point P

Question 7. The curve

$$y = \frac{x^2 - 2x}{x^2 - 8x + 16}$$

is shown in the diagram below. The local minimum is labelled Q



a) By considering where the line $y = k$ intersects the curve, show that

$$32k + 4 \geq 0$$

b) State the values that y can be.
c) Hence, find the y-coordinate of the point Q

Question 8. A graph has equation

$$y = \frac{x + 2}{x^2 + 5}$$

a) Explain why the graph has no vertical asymptote.
b) Find the equation of the horizontal asymptote.
c) Show that, if the line $y = k$ intersects the graph, the x-coordinates of the points of intersection satisfy the equation

$$kx^2 - x + (5k - 2) = 0$$

d) Show that this equation has real roots if

$$-\frac{1}{10} \leq k \leq \frac{1}{2}$$

e) Hence, find the coordinates of the two stationary points on the curve.

Question 9.

a) Prove that

$$-3 \leq \frac{12x}{x^2+4} \leq 3$$

b) Hence, sketch

$$\mathrm{f}(x) = \frac{12x}{x^2+4}$$

c) Given that $\mathrm{g}(x) = \dfrac{1}{\mathrm{f}(x)}$ explain why

$$-\frac{1}{3} \leq \mathrm{g}(x) < 0 \text{ and } 0 < \mathrm{g}(x) \leq \frac{1}{3}$$

Question 10. Sketch each of the following curves on separate axes. Label any intersections with the axes, asymptotes and (local) maximum or (local) minimum points. You should find (local) maximum and (local) minimum points by considering where the line $y = k$ intersects the graph.

a) $y = \dfrac{x^2+4}{x^2-1}$

b) $y = \dfrac{x^2-4x+4}{x^2+4x}$

c) $y = \dfrac{x^2+8x+16}{x^2+2x+2}$

d) $y = \dfrac{x^2+4x}{x^2+2}$

e) $y = \dfrac{x^2-4x-5}{x^2+4x}$

f) $y = \dfrac{x^2-4x+2}{x^2-4x+3}$

Question 11. Find any asymptotes for each of the graphs of the following curves:

a) $y = \dfrac{2x^2 + x + 1}{x^2 + 2x + 5}$

b) $y = \dfrac{3x^2 - 1}{x^2 + 2x + 3}$

c) $y = \dfrac{x^2 + 9}{4x^2 - 8x + 4}$

d) $y = \dfrac{5x^2 + 14x - 3}{2(2x^2 + 7x - 15)}$

Question 12. A curve is given by the equation

$$y = \frac{ax^2 + bx + c}{x^2 - d}$$

where a, b, c and d are positive constants. Write down the equations of its asymptotes in terms of the constants.

Question 13. A curve has equation

$$y = \frac{4x^2 + 4x + 11}{x^2 - x + 1}$$

It is known that the curve has a single maximum point and a single minimum point.

a) Write down the equation of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

b) Show that

$$(k - 20)(3k - 8) \leq 0$$

c) Hence find the coordinates of the stationary points of the curve and identify which is the minimum and which is the maximum. Give a reason for your answer.

d) Hence, sketch the curve.

Question 14. A curve has equation

$$y = \frac{2x^2 - 4x + 9}{x^2 - 4}$$

for $x \neq \pm p$ where p is a constant.

a) State the value of p

b) Write down the three equations of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

c) Show that

$$(4k - 7)(k + 2) \geq 0$$

d) Hence find the coordinates of the stationary points of the curve and identify which is the local minimum and which is the local maximum.

e) Hence, sketch the curve.

Question 15. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 - 8x + 1}{3x^2 + 1}$$

a) Write down the equation of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

b) Show that

$$(3k + 5)(k - 3) \leq 0$$

c) Hence find the coordinates of the stationary points of the curve, stating which is the maximum and which is the minimum.

d) Sketch the graph of $y = \mathrm{f}(x)$

Question 16. For each of the following, state the equation of the single asymptote of the graph and, without using Calculus, find the coordinates of the stationary point of the curve.

a) $y = \dfrac{1}{2x^2 + 3}$

b) $y = \dfrac{1}{3(x^2 + 2x + 2)}$

Question 17. A curve is given by $y = \mathrm{f}(x)$ and it is known to have a local maximum at (a, b) where a and b are constants.
A second function is defined to be

$$\mathrm{g}(x) = \frac{1}{\mathrm{f}(x)}$$

It is given $\mathrm{f}(x)$ and $\mathrm{g}(x)$ have the same stationary point. State the type of stationary point and any restrictions on this occurring.

Question 18. Two curves C_1 and C_2 are given by

$$\mathrm{f}(x) = x^2 - 4x + 5$$

and

$$y = \frac{1}{\mathrm{f}(x)}$$

respectively.

a) Find the minimum point on the graph of curve C_1
b) Sketch the graph of $y = \mathrm{f}(x)$
c) Without using Calculus, determine the single maximum point on the graph of curve C_2
d) Hence, state the equation of the line that is a tangent to both C_1 and C_2

Question 19. A curve C_1 has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x^2 - 8x + 7}{x^2 + 4x - 6}$$

The line $y = k$ intersects the curve C_1

a) Show that

$$(k + 1)(2k - 1) \geq 0$$

b) Hence find the coordinates of the stationary points of the curve C_1 and identify their type.

A second curve C_2 is given by

$$y = \frac{1}{\mathrm{f}(x)}$$

c) Show that the graph of C_2 has no vertical asymptotes.

d) State the equation of the line that is a tangent to both C_1 and C_2

Question 20. A curve C is given by the equation

$$y = \frac{6 - x^2}{x^2 + kx + 2}$$

where k is a constant.

It is known that the graph of the curve only has one asymptote.

a) Write down the equation of the asymptote of the graph of the curve.

b) Find the possible values of k

c) Find where the graph of the curve C intersects the coordinate axes.

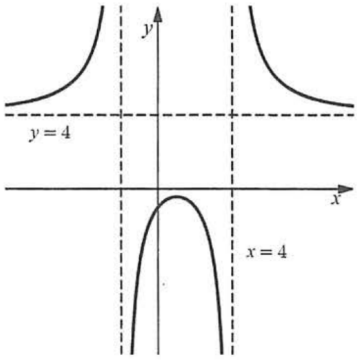
d) Sketch the graph of the curve given by

$$y = \frac{6 - x^2}{x^2 + 2}$$

Question 21. The curve C has equation

$$y = \frac{ax^2 + 4bx + 2a}{x^2 + bx - 2a}$$

where a and b are integers and is shown in the diagram below.



One of the vertical asymptotes is $x = 4$
The horizontal asymptotes is $y = 4$

a) State the value of a
b) Find the value of b
c) State the equation of the second vertical asymptote of the curve.
d) Find the coordinate of the intercept of the curve with the y-axis.
e) Without using Calculus, find the coordinates of the local maximum point.

Question 22. A curve is given $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + bx + c}{x^2 + px + q}$$

where a, b, c, p and q are integers.
It is known that:

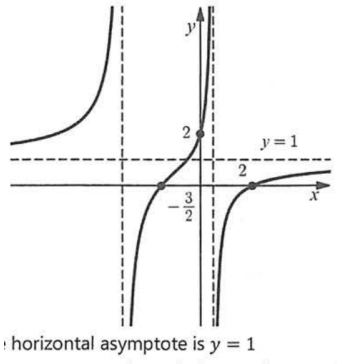
- There is a single asymptote at $y = 2$
- It crosses the y-axis at -1
- It crosses the x-axis at $-1 + \sqrt{\frac{3}{2}}$ and $-1 - \sqrt{\frac{3}{2}}$
- It has a minimum at $\left(-\frac{1}{2}, -2\right)$

a) Find $\mathrm{f}(x)$
b) Given that the curve also has a maximum, without using Calculus find the coordinates of this maximum point.
c) Sketch the graph of this function.

Question 23. A curve is given $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + bx + c}{px^2 + qx + r}$$

where a, b, c, p, q and r are integers. The graph of $y = \mathrm{f}(x)$ is shown in the diagram below.



horizontal asymptote is $y = 1$

The horizontal asymptote is $y = 1$

The curve passes through the y-axis at $y = 2$ and the x-axis at $x = -\frac{3}{2}$ and $x = 2$

a) State the relationship between a and p

b) State the relationship between c and r

It is given additionally that $a = 2$ and that one of the vertical asymptotes is $x = -3$

c) Find $\mathrm{f}(x)$

d) Find the other vertical asymptote.

Question 24. The curve C has equation

$$y = \frac{x^2 + p}{x^2 - q^2}$$

where p and q are positive integers.

Without using Calculus, find the range of values that y can take in terms of p and q

Question 25. The curve C has equation

$$y = \frac{x^2 - px + p}{x^2 + q}$$

where p and q are positive integers.

a) State the asymptotes of curve C

b) Without using Calculus, show that the graph does not exist for

$$y < \frac{p + q - \sqrt{p^2 - 2pq + q^2 + p^2 q}}{2q}$$

and

$$y > \frac{p + q + \sqrt{p^2 - 2pq + q^2 + p^2 q}}{2q}$$